

1) určete VP, průsečík P

$$\begin{array}{ll} p: x = 1 - 2t & q: x = 4 + r \\ y = 8 + 3t & y = 3 - 2r \\ z = 6 - t & z = 10 + 3r \end{array}$$

$$\vec{s}_p = (-2; 3; -1) \quad \vec{s}_q = (1; -2; 3)$$

$$\left. \begin{array}{l} -2 = 1t \\ 3 = -2t \Rightarrow -\frac{3}{2} = t \\ -1 = 3t \end{array} \right\} -2 \neq -\frac{3}{2} \Rightarrow \begin{cases} \text{různoběžky} \\ \text{mimoběžky} \end{cases}$$

$$1 - 2t = 4 + r \Rightarrow r = -3 - 2t$$

$$8 + 3t = 3 - 2r$$

$$6 - t = 10 + 3r$$

$$8 + 3t = 3 - 2(-3 - 2t)$$

$$6 - t = 10 + 3(-3 - 2t)$$

$$8 + 3t = 3 + 6 + 4t$$

$$6 - t = 10 - 9 - 6t$$

$$8 + 3t = 9 + 4t \Rightarrow t = -1$$

$$6 - t = 1 - 6t \Rightarrow t = -1 \quad \left. \begin{array}{l} \Rightarrow t = -1 \\ \Rightarrow t = -1 \end{array} \right\} -1 = -1 \Rightarrow \text{různoběžky}$$

$$x = 1 + 2 = 3$$

$$y = 8 - 3 = 5$$

$$z = 6 + 1 = 7$$

$$P = [3; 5; 7]$$

2) VP

$$P = [1; 2; -1]$$

$$\begin{aligned} p: x &= 1+3t & q: x &= -3r \\ y &= 2-t & y &= 1+r \\ z &= -1+t & z &= 5-r \end{aligned}$$

$$\vec{s}_p = (3; -1; 1) \quad \vec{s}_q = (-3; 1; -1)$$

$$\left. \begin{aligned} 3 &= -3t \Rightarrow t = -1 \\ -1 &= t \\ 1 &= -t \end{aligned} \right\} -1 = -1 = -1 \Rightarrow \begin{cases} r=0 \vee h=0 \\ t=0 \wedge z=2 \end{cases}$$

$$\left. \begin{aligned} 1 &= -3r \Rightarrow r = -\frac{1}{3} \\ 2 &= 1+r \Rightarrow r = 1 \\ -1 &= 5-r \Rightarrow r = 6 \end{aligned} \right\} -\frac{1}{3} \neq 1 \neq 6 \Rightarrow p \parallel q \wedge p \neq q$$